Avishi Mittal

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Education

Thapar Institute of Engineering and Technology **Sep 2022 – May 2026**
Bachelor of Engineering in Computer Engineering

The British School, Mohali **May 2020 – Sep 2022**
PCM, CBSE

Relevant Coursework

- Data Structures
- Operating Systems
- Machine Learning
- Database Management
- Data Science
- Software Engineering
- Artificial Intelligence
- Computer Networks

Experience

a21.ai **Jan 2026 – Present**
AI Engineer *New Delhi, India*

- Led a team of 4 to build **Text-to-SQL** system with **96% accuracy** enabling natural language queries on relational database
- Delivered modular components in **production AI pipeline** with **end-to-end ownership** from requirements to integration
- Refined **30+ prompts** using **few-shot** and **chain-of-thought prompt engineering** techniques to improve accuracy
- Architected an **agentic pipeline** using **LangGraph**, **Neo4j knowledge graph traversal** using **hybrid retrieval**
- Implemented **human-in-the-loop (HITL)** workflows with LangGraph interrupts, PostgreSQL-backed state persistence
- Instrumented the LLM pipeline with **Langfuse** for **prompt versioning, tracing, and cost and latency observability**

Shiksha Sathi **Jun 2025 – Aug 2025**
Full Stack Developer Intern *Remote*

- **Refactored 10+ web pages** through code reviews to improve layout, responsiveness, and accessibility.
- Optimized website performance and backend APIs, boosting mobile **PageSpeed SEO score by 130%**

Projects

Research-RAG | *Python, FastAPI, ChromaDB, SQLite* **Link**

- Designed **hybrid RAG pipeline (BM25 + embeddings)** with **ChromaDB vector indexing** for semantic search
- Enabled **configurable weighting** and **metadata filtering** across dense and sparse retrieval methods for context retrieval
- Integrated **Gemini LLM API** for context-grounded answers with **source citation** and relevance ranking

Smart EHR System | *Python, FastAPI, React, FHIR R4, LSTM* **Link**

- Developed a **Smart EHR platform**, delivering a scalable **3-tier system** for clinical data and operational workflows.
- Engineered a **FHIR R4-compliant backend** supporting **8+** healthcare resource types, enabling interoperable workflows
- Trained **multi-modal LSTM** on **18,000+ longitudinal patient visits** from ADNI dataset, achieving **0.95 R^2**

Bug Tracker | *React, Node.js, MongoDB, JWT, REST APIs* **Link**

- Built a **full-stack Bug Tracker** for simplifying issue resolution workflow, deployed on **Vercel** and **Render**
- Established **role-based** workflows for 3 user personas, with **JWT auth, authorization, validation, and rate limiting**
- Developed end-to-end **CRUD** for bugs, projects, and users, with lifecycle tracking and public bug intake

Technical Skills

- **Languages**: C/C++, Python, JavaScript, HTML/CSS, SQL
- **Libraries And Frameworks**: Langchain, Langgraph, React.js, Node.js, FastAPI
- **Databases**: PostgreSQL, MongoDB, ChromaDB
- **Tools**: Langfuse, pgAdmin, Postman, Git/GitHub, VS Code, Codex, Claude Code, Open AI API

Achievements/Extracurricular

- Cleared **AWS Certified Solutions Architect** - Associate certification [**Link**]
- Organized technical events and workshops for Microsoft Learn Student Chapter; co-led planning of **MLH Sponsored** Makeathon with **200+ participants**